

William Sander

Software Engineering | Applied Machine Learning

Chicago, IL · Open to Relocation · Phone: (802) 307-6130 · Email: workwilliamsander@gmail.com

Website: willchristophersander.github.io · Github: github.com/willchristophersander · LinkedIn: linkedin.com/in/willsander

Summary

Systems-oriented Software Engineer (BS May 2026) specializing in Applied ML and production systems. Proven track record of architecting end-to-end solutions, from multilingual deep learning models to large-scale infrastructure automation for the U.S. Census Bureau. Experienced in Python, C++, and PyTorch; targeting roles in Chicago or the SF Bay Area.

Education

University of Vermont

Burlington, VT

B.S. Computer Science

Expected May 2026

Relevant Coursework: Advanced Machine Learning, Advanced Programming (C++), Algorithms, Data Structures, Database Systems, Operating Systems

Hackathons & Contests: UVM CS Fair 2025 & 2026, UVM Hackathon 2026

Projects

AI Speech Emotion Recognition System

UVM, Advanced Machine Learning

PyTorch / Python / Deep Learning / ML Evaluation

[GitHub](#)

- Built a CRNN combining CNNs and BiLSTMs with attention, achieving **81.29% accuracy and 83.63% F1** (4-class) and **76.76% accuracy and 82.80% F1** (6-class) across **8 multilingual datasets** (~25.8k samples)
- Added **distance-weighted loss** using arousal-valence theory; benchmarked against classical ML baselines (253-feature extraction, ensemble models)
- Built modular data pipeline with interpretability tools (attention maps, Grad-CAM); deployed live inference web app

C++ Canvas Raster Graphics Editor

UVM, Advanced Programming

C++ / OpenGL / Command Pattern / NumPy

[Video Demo](#) · [GitHub](#)

- Built real-time raster graphics engine with **7 tools**, multi-layer RGBA buffers, and **diff-based undo/redo**; custom rasterization, per-pixel alpha compositing, and zero-copy buffer reuse keep memory under **140 MB**
- Implemented OpenGL pipeline with **lazy CPU-to-GPU texture updates**, achieving **60 FPS** at **800×600**; integrated Gaussian, Sobel, and unsharp filters via **Python C API + NumPy** with direct C++↔Python buffer access and no copy overhead

E-Ink iCloud Calendar Display

IoT / OOP / Python / 3D Printing / FreeCAD / Hardware

[GitHub](#)

- Built an IoT E-Ink calendar device integrating iCloud and public holiday APIs; unified ICS/JSON event streams into a custom rendering pipeline
- Designed modular OOP architecture with custom layout engine; 3D-printed enclosure in FreeCAD

Work Experience

IT Clerk — Automation & Systems Software Engineering

Burlington, VT

U.S. Census Bureau

Feb 2020–Nov 2020

- Took ownership of software development within the IT team, engineering and maintaining tools to diagnose and resolve large-scale device and configuration failures across distributed census infrastructure — sustaining uptime across **hundreds of enumerators** and **3,000+ mobile devices**
- Built validation and consistency software to detect and correct device custody and configuration issues across distributed infrastructure, improving reliability during large-scale field operations
- Identified manual processing bottlenecks and built production automation in **Python, C++, and SQL**, eliminating **30+ hours/week** of reconciliation work (**1,000+ hours annually**)
- Architected and deployed production software under real-time operational constraints during the COVID-19 census extension
- Supported authentication, access control, and compliance across **3,000+ mobile devices** in coordination with regional IT teams

Skills

Languages: Python, C++, C, Java, R, SQL

ML & Data: PyTorch, scikit-learn, NumPy, Flask

Tools & Systems: Git, Docker, Linux, REST APIs, VSCode